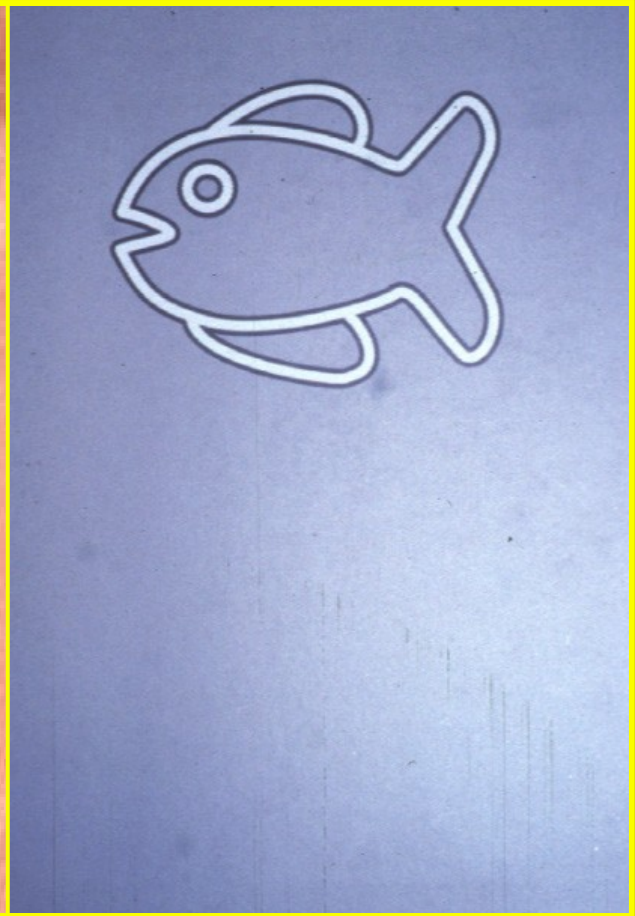
CHILDHOOD STRABISMUS

1. **Examination**
2. **Esotropia**
   - Essential infantile esotropia
   - Refractive accomodative esotropia
   - Non-refractive esotropia
3. **Exotropia**
   - Constant exotropia
   - Intermittent exotropia
4. **Special syndromes**
   - Duane syndrome
   - Brown syndrome
   - Double elevator palsy
   - Möbius syndrome
5. **Alphabet patterns**
   - ‘V’ pattern deviation
   - ‘A’ pattern deviation

Visual acuity tests in preverbal children

‘Hundreds and thousands’ sweet test



Preferential looking with Cardiff cards

Visual acuity tests in verbal children

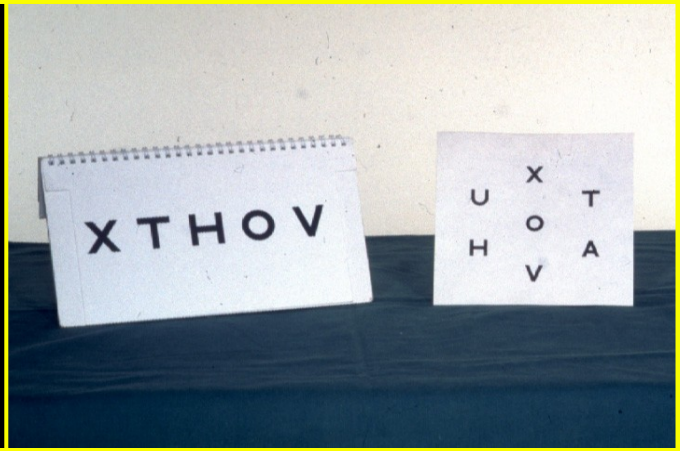
At age 2 years (naming pictures)

Kay single picture

Multiple pictures

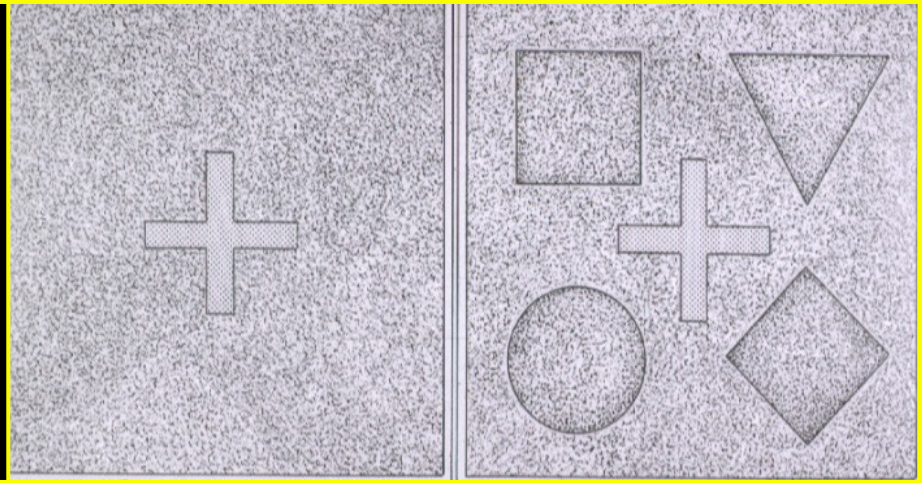
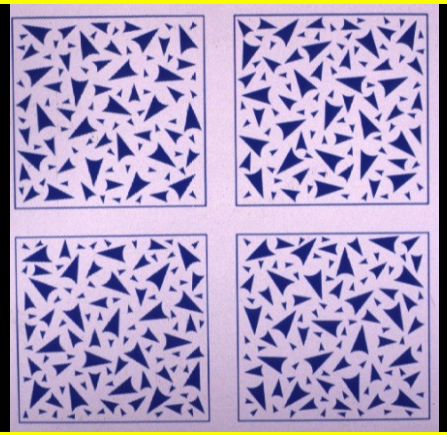
At age 3 years (matching tests)

Sheridan-Gardiner



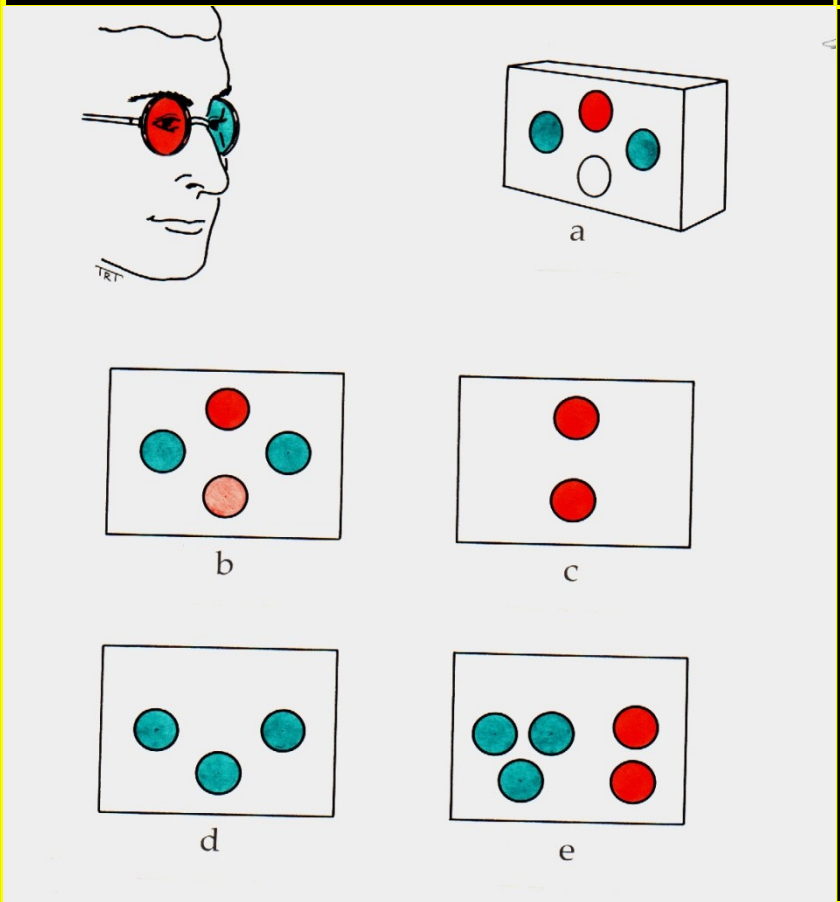
Sonksen-Silver

Tests for stereopsis

Titmus	TNO random dot test
	
· Polaroid spectacles · Figures seen in 3-D	· Red-green spectacles · ‘Hidden’ shapes seen
Frisby	**Lang**
	
· No spectacles · ‘Hidden’ circle seen	· No spectacles · Shapes seen

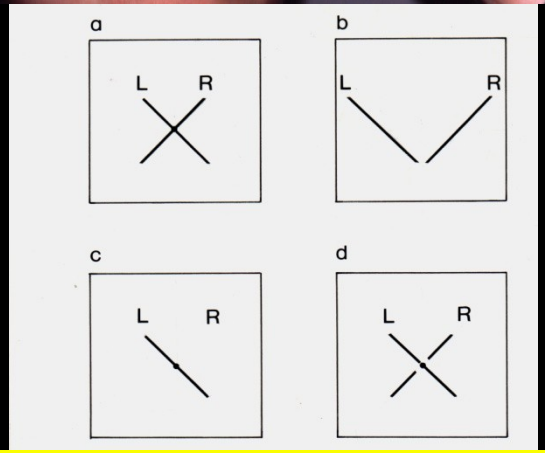
Tests for sensory anomalies

Worth four-dot test



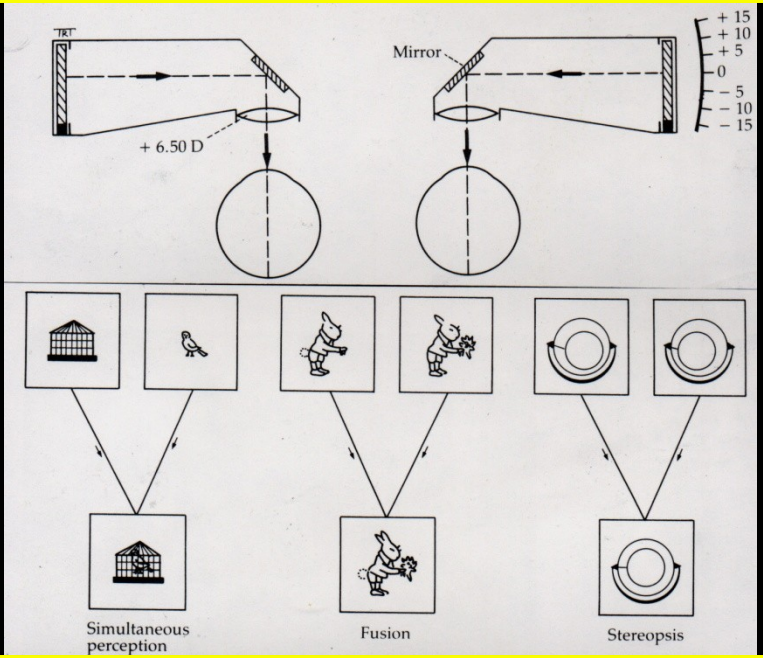
a - Prior to use of glasses
b - Normal or ARC
c - Left suppression
d - Right suppression
e - Diplopia

Bagolini striated glasses



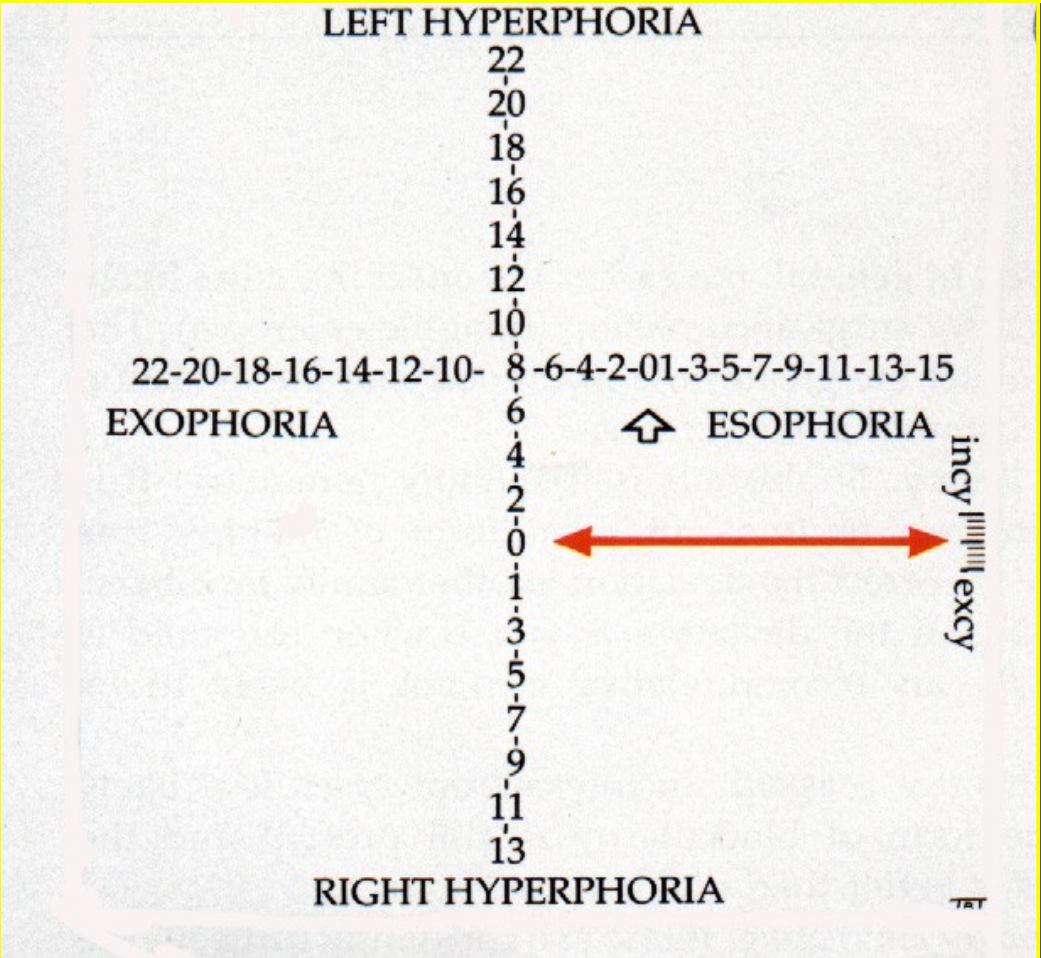
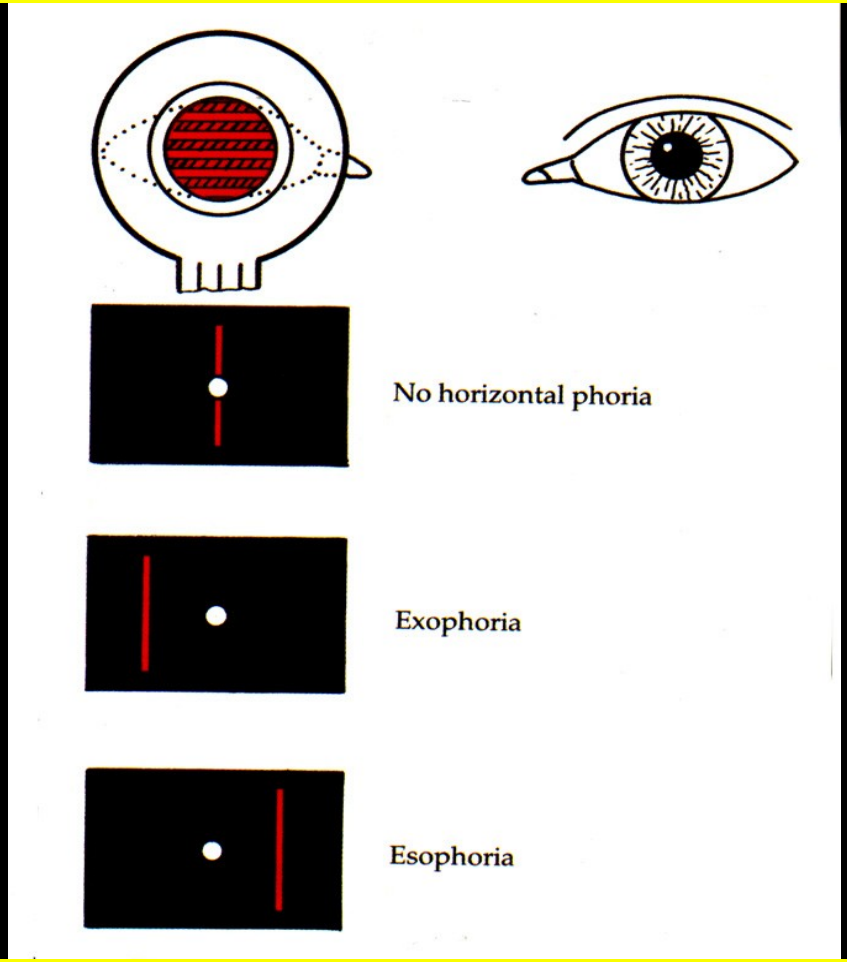
a - Normal or ARC
b- Diplopia
c - Suppression
d - Small suppression scotoma

Synoptophore



- Grading of binocular vision
- Detection of suppression and ARC
- Measurement of angle
- Measurement of fusional amplitudes

Dissimilar image tests

Maddox wing	Maddox rod
	
· Dissociates eyes for near fixation (1/3 m) · Measures heterophoria	· White spot converted into red streak · Cannot differentiate tropia from phoria

Hirschberg test

- Rough measure of deviation
- Note location of corneal light reflex
- 1 mm = 7° or 15△

Reflex at border of pupil = 15 △

Reflex at limbus = 45 △

Pseudo-deviations

Pseudo-esotropia	Pseudo-exotropia
• Epicanthic folds • Short interpupillary distance • Negative angle kappa	• Wide interpupillary distance • Positive angle kappa

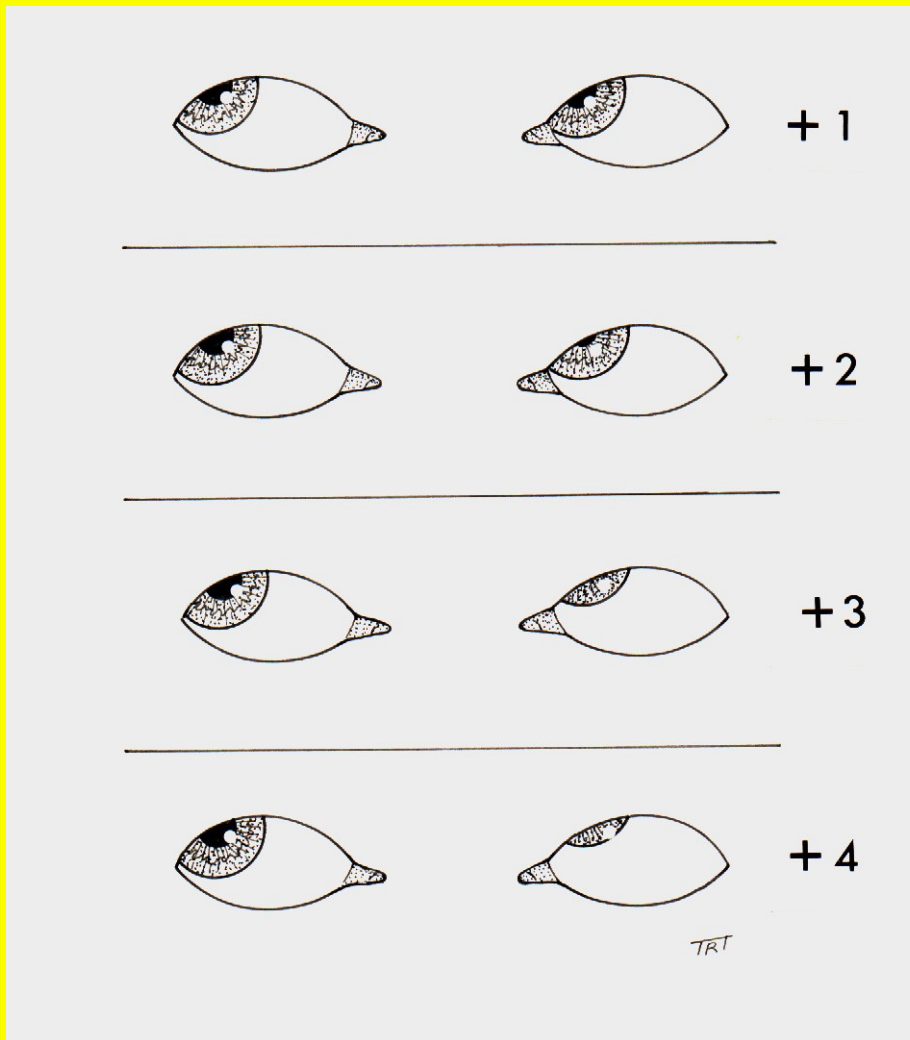
Cover tests

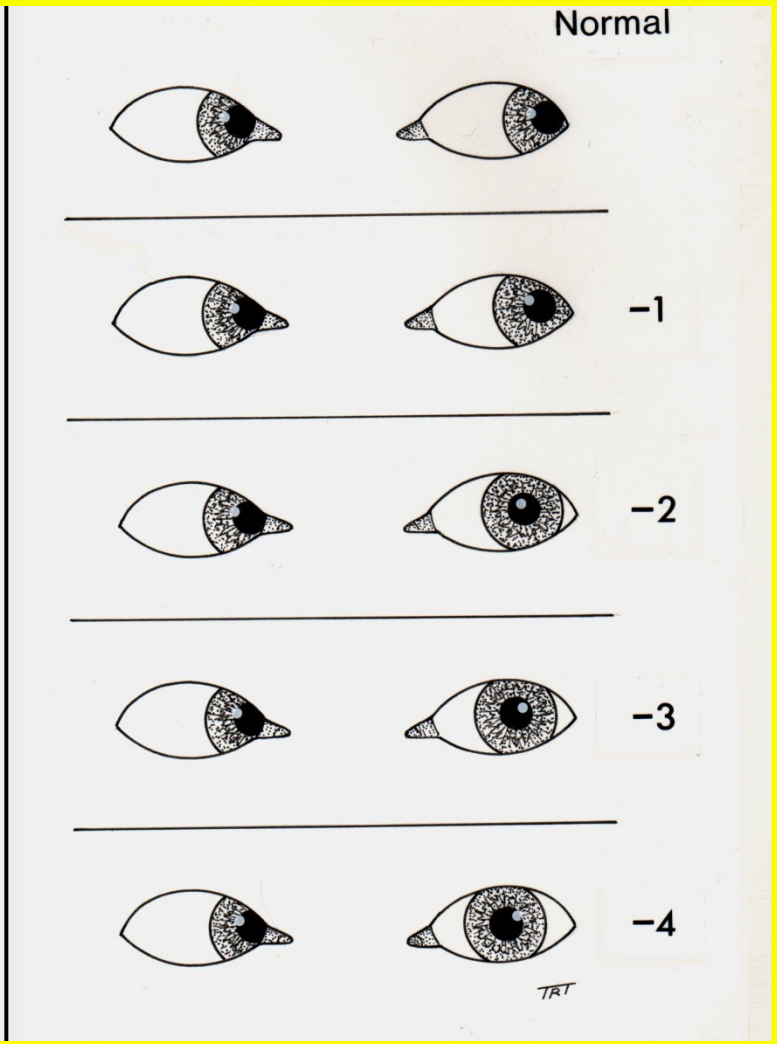
- Cover test detects heterotropia
- Uncover test detects heterophoria
- Alternate cover test detects total deviation

- Prism cover test measures total deviation

Motility tests

- Tests versions and ductions
- Grades under/overaction

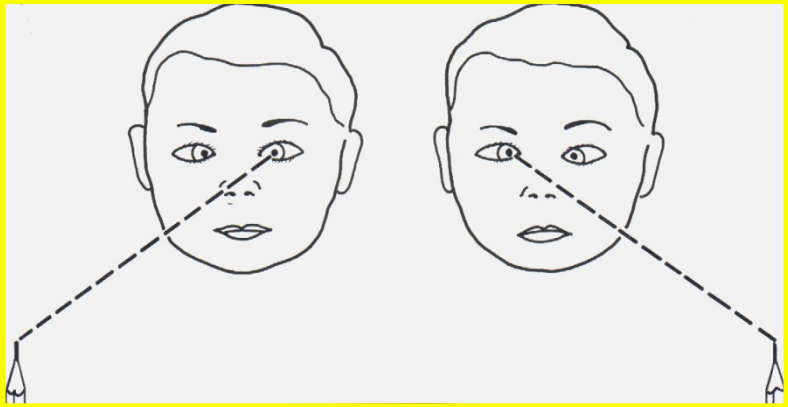




Left inferior oblique overaction	Left lateral rectus underaction

Essential infantile esotropia

Presents within first 6 months

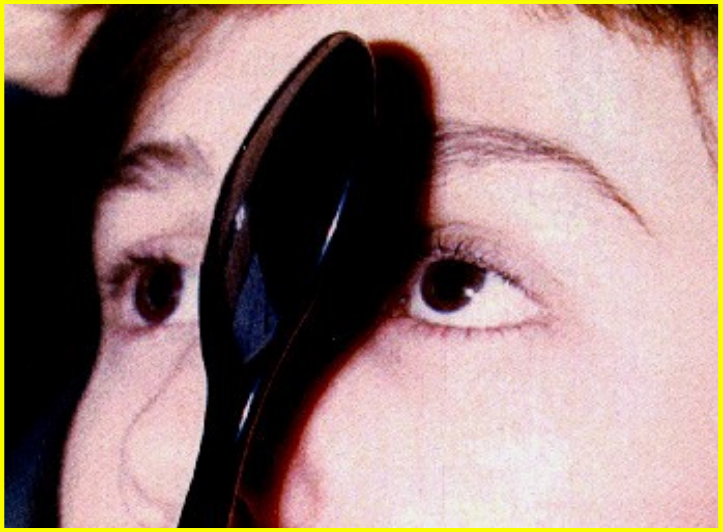
Signs	
· Angle large and stable · Nystagmus in some cases · Normal refraction for age · Poor potential for BSV · Amblyopia in about 30%	 · Cross fixation

Management of essential infantile esotropia

- Correct amblyopia if present
- Surgery before age 12 months

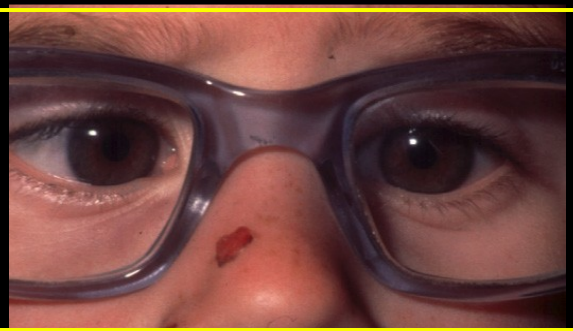
- Bilateral medial rectus recessions
- Ideal alignment within 10 Δ

Subsequent problems

Inferior oblique overaction	Dissociated vertical deviation	Microtropia
		
· Most common onset 2 years · Usually eventually bilateral	· Up-drift with excyclodeviation of eye under cover · When cover removed affected eye moves down	· Very small angle - may not be detectable on cover testing · Central suppression scotoma

Refractive accommodative esotropia

- Presents between 18 months - 3years
- Initially intermittent
- Normal AC/A ratio
- Excessive hypermetropia

Fully accommodative	Partially accommodative
Esotropia greater for near	Straight for distance
	
Straight for distance and near	Esotropia for near

Non-refractive accommodative esotropia

- Presents between 18 months to 3years
- High AC/A ratio
  - due to increased AC (convergence excess)
  - due to decreased A (hypoaccommodative)
- No significant refractive error

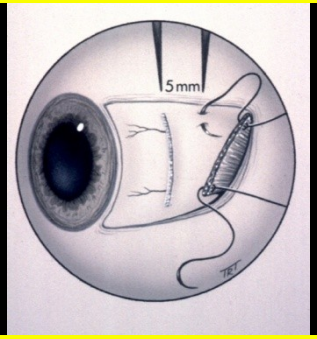
Signs	
Straight for distance	Esotropia for near

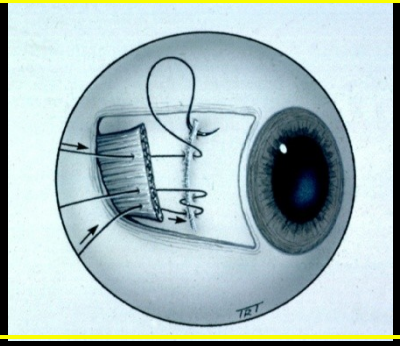
Management of accommodative esotropia

Refraction - prescribe full cycloplegic refraction under age 6 years

Treatment of amblyopia

Surgery - if spectacles do not fully correct deviation





Recession	Resection

Constant exotropia

Congenital	Sensory
• Presents at birth • Large angle • Alternating fixation • Normal refraction for age	• Disruption of binocular reflexes by acquired lesions, such as cataract

Consecutive - follows previous surgery for esotropia

Intermittent exotropia

Signs

- Presents - usually prior to 5 years
- Usually alternating (amblyopia uncommon)
- Treatment - surgery

Basic

- Angle greater for near

Convergence weakness

- Angle greater for near
- May be associated with myopia

Divergence excess

- Angle greater for distance
- May be true or simulated

Duane syndrome

- Bilateral in about 20%
- On attempted adduction - retraction of globe and narrowing of palpebral fissure
- On attempted abduction - opening of palpebral fissure and normal globe position

Left type I (left)

Adduction- normal or mildly limited

Primary position - straight or mild esotropia

Abduction - limited or absent

Type II

- Abduction - normal or mildly limited
- Adduction - limited
- Primary position - straight or mild exotropia

Type III (left)

Abduction - limited

Primary position - straight or mild esotropia

Adduction - limited

Brown syndrome (right)

Normal elevation in abduction	Straight in primary position	Limited elevation in adduction

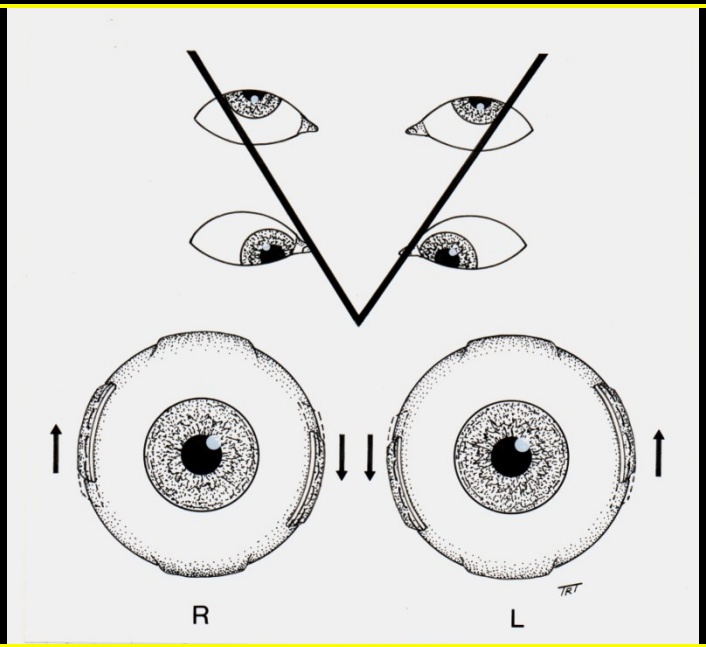
Double elevator palsy (right)

Unilateral elevation failure in all positions

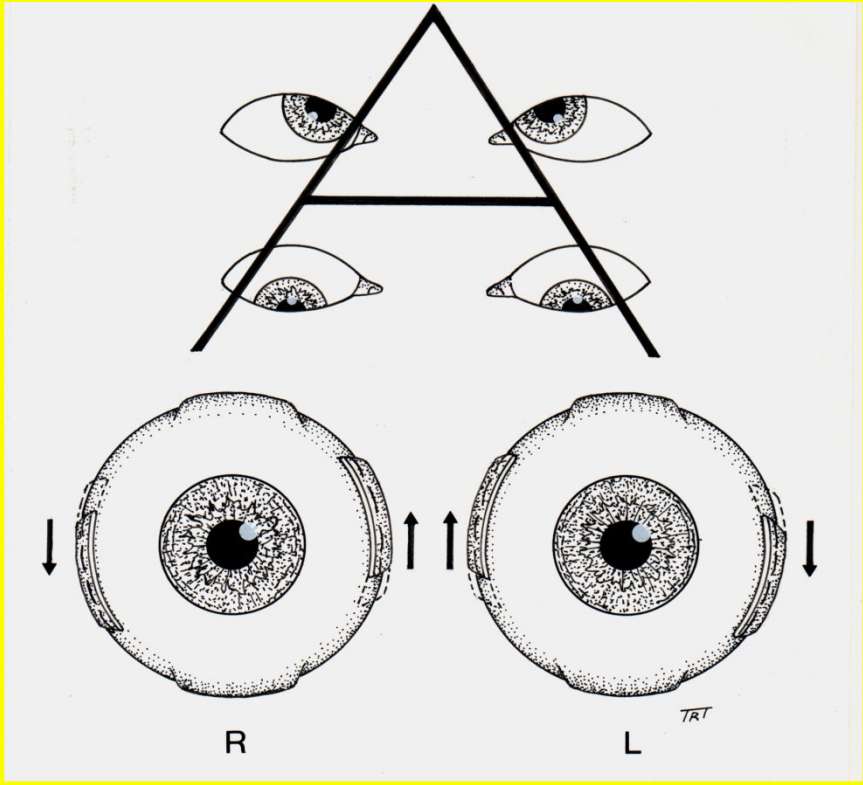
Möbius syndrome

Signs		
• Bilateral sixth nerve palsies - patient looking left	• Primary position - 50% straight, 50% esotropic • Horizontal gaze palsy in 50%	• Bilateral, usually asymmetrical facial palsies sparing lower face • Paresis of 9th and 12th cranial nerves

‘V’ pattern deviation

Signs	Treatment
	
· Difference between up- and downgaze is 15△ or more	**‘V’ pattern esotropia** · Bilateral medial rectus recessions + downward transposition **‘V’ pattern exotropia** · Bilateral lateral rectus recessions + upward transpositions

‘A’ pattern deviation

Signs	Treatment
	
· Difference between up- and downgaze 10 Δ or more	**‘A’ pattern esotropia** · Bilateral medial rectus recessions + upward transposition **‘A’ pattern exotropia** · Bilateral lateral rectus recessions + downward transposition